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Cluster 6

The Birthday Paradox: A Simple and Surprising Probability Problem

I. Introduction

The birthday problem is a probability problem that deals with the probability of at least one pair of people in a group of n people that share the same birthday. For example, one could ask,

"How many people are required such that it is more likely than not that at least two people in the group share the same birthday?"

To many, this might seem like a large number of people are required. However, this is incorrect, as only 23 people are required to have a probability of around 50.7%. This goes against the intuition that it would take 183 people, which would be the answer for the question if there was a set birthday that needed to be matched. However, instead, in this birthday problem, every birthday is randomized.

II. Solving the Problem

In order to solve the problem, one must find the probability of the complement, . By subtracting this probability from 1, one can find the probability of at least one pair having the same birthday. For example, the probability of 40 people having at least one of the same birthday goes as follows:

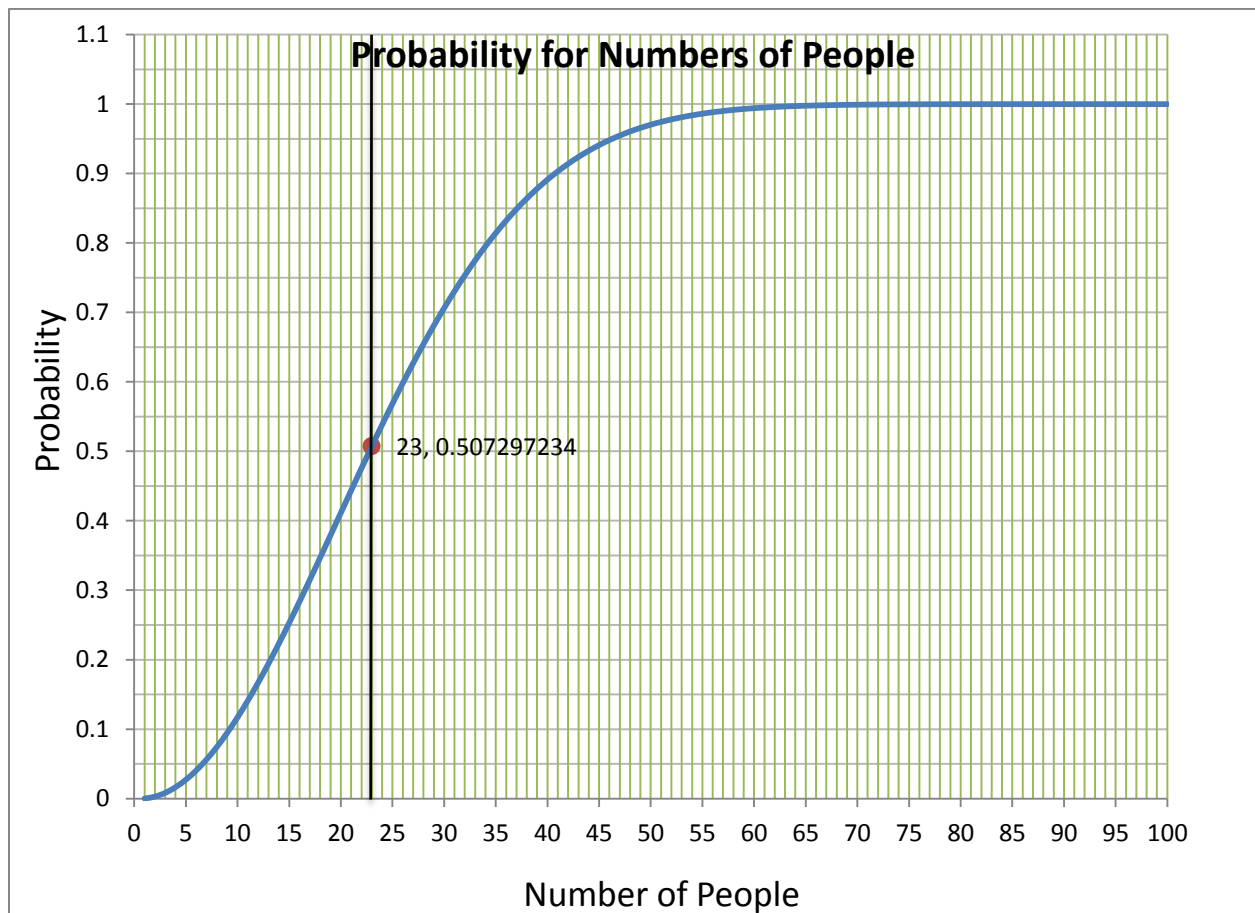
$$\begin{aligned}P(A) &= 1 - P(\bar{A}). \\P(\text{At least 1 same birthday}) &= 1 - P(\text{No same birthdays}). \\&= 1 - \left(\frac{365}{365} \times \frac{(365 - 1)}{365} \times \dots \times \frac{(365 - 39)}{365} \right). \\&\approx 0.89123.\end{aligned}$$

For this example, the process that is being followed is finding the number of ways for a person to not share a birthday with any of the other people. This could be thought of as putting 40 balls into 365 slots in order so that each slot can have at most one ball, divided by 365, the

total number of possible locations to choose from. The first choice for the birthday could be any of the 365, and then second could have 364, and so on. This equation could be simplified, though, such as the following:

$$\begin{aligned}
 1 - \left(\frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365 - n + 1}{365} \right) &= 1 - \left[\frac{365!}{365^n \times (365 - n)!} \right] \\
 &= 1 - \left(\frac{n! \times C(365, n)}{365^n} \right) \\
 &= 1 - \left(\frac{P(365, n)}{365^n} \right)
 \end{aligned}$$

$365 \times 364 \cdots \times (365 - n + 1)$ can be simplified into $\frac{365!}{(365-n)!}$, which then can be simplified into $n! \times C(365, n)$, and then simplified again into $P(365, n)$ while the denominator simplifies into 365^n . Using this, one can create a graph and find the probability for every number of people, such as the one on the next page.



As one can see, the graph is much like a logistic equation, except for the fact that at 366 people, the probability will actually reach 1, due to the pigeonhole principle, where if the number of pigeons exceeds the number of holes, then there must be at least one pair of pigeons in one of the holes. In reality, the graph should be a staircase diagram, but we shall work with the assumption that people can be split into fractions. As seen on the graph, the probability reaches over .5 at the point (23, 0.507297234), or 23 people. Therefore, it is more likely that not that for all the students and staff in Cluster 6 of COSMOS, including Residential and Teaching Assistants as well as professors, at least 2 people share the same birthday. The probability increases at a surprising rate, with the probability being over 0.9 with 41 people and being over 0.99 with 57 people.

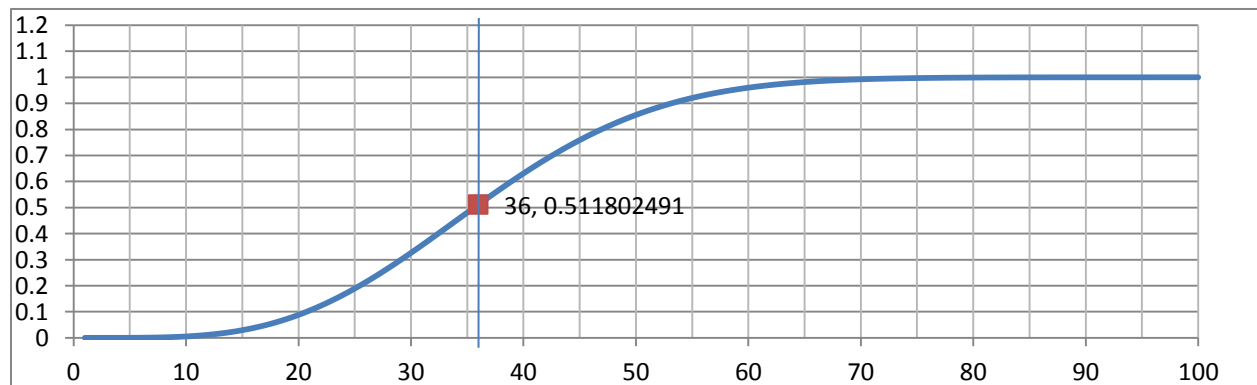
III. Variations

A variant of the birthday problem could be finding the probability that there are at least 2 pairs of people sharing the same birthday instead of just one. Using complements again, one simply adds the probability of getting only one pair of birthdays to the probability of having no birthdays as shown above, and subtracts this whole value by 1. For example:

$$P(\text{at least 2 pairs}) = 1 - P(\text{no pairs} + 1 \text{ pair})$$

$$\begin{aligned} P(n) &= 1 - \left[\frac{P(365, n)}{365^n} + \frac{C(n, 2) \times P(365, n - 1)}{365^n} \right] \\ &= 1 - \left(\frac{P(365, n) + C(n, 2) \times P(365, n - 1)}{365^n} \right). \end{aligned}$$

Since there are n people being selected, and only 2 are chosen to be the birthday pair, the number of arrangements is $C(n, 2)$. Then, there are $P(365, n - 1)$ ways to arrange the remaining people, as the pair can be counted as one person.



This graph is much like the original birthday problem graph, except with the probability being likely at 36 people.

Say, though, that one replaces the birthdays with the last four digits of social security numbers. What would be the probability, then, that at least one pair of n people share the same last four digits of a social security number?

Instead of 365 days to choose from, there are 9999 different numbers to choose from. Assuming that the chosen social security numbers are random, the problem can be solved exactly like the last problem.

$$\begin{aligned}
 P(n) &= 1 - P'(n) \\
 &= 1 - \left(\frac{9999}{9999} \times \frac{9998}{9999} \times \cdots \frac{9999 - n + 1}{9999^n} \right) \\
 &= 1 - \left[\frac{9999!}{9999^n \times (9999 - n)!} \right] \\
 &= 1 - \left[\frac{n! \times C(9999, n)}{9999^n} \right] \\
 &= 1 - \left[\frac{P(9999, n)}{9999^n} \right]
 \end{aligned}$$

This problem is obviously much like the original birthday problem. However, a problem arises when attempting to graph it, as the numbers get too far for even Microsoft Excel to handle. Instead, an approximation of e can be used in order for the graph to be mostly accurate. In the Taylor's series expansion of e , the value of e^x is written as:

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots$$

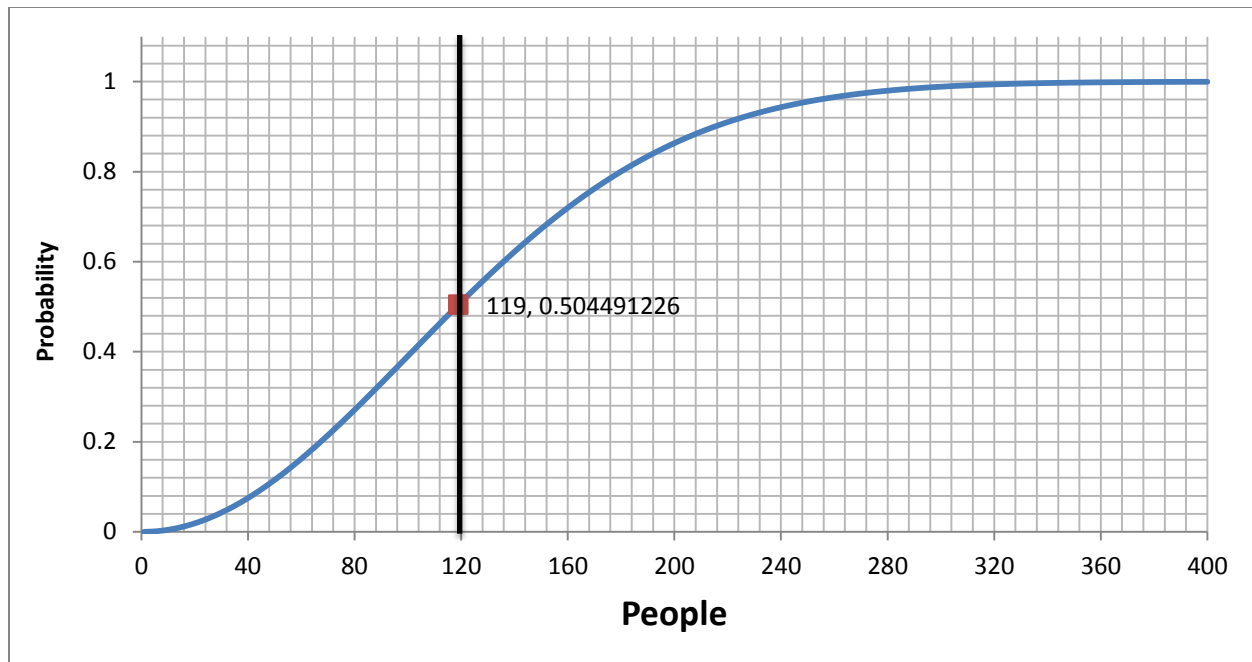
However, when $x \ll 1$, it can be approximated as:

$$e^x = 1 + x.$$

In order to fit the social security number question, one can replace x with $\frac{-(n-1)}{9999}$ so that it matches the original equation. The equation then follows the format:

$$\begin{aligned}
 P(n') &= 1 \times e^{\frac{-1}{365}} \times e^{\frac{-2}{365}} \cdots e^{\frac{-(n-1)}{9999}}. \\
 &= e^{-(1+2+\cdots+(n-1))/365}. \\
 &= e^{[-n(n-1)/(2 \times 365)]}.
 \end{aligned}$$

The graph is then like the following:



At 119 people, the chance of a pair having a duplicate last four digits of a social security number is 50.45%.

IV. Application

The mathematics of the birthday problem is used in the birthday attack, a brute-force cryptographic attack against hash function problems. These hash functions are either well-defined procedures or mathematical functions which convert large amounts of data into a small, single-integer datum, called a hash, hash value, hash code, hash sum, or checksum. This is done in order to speed up the looking up of items in a database. However, since there are a limited amount of hashes, there can be many collisions of hashes. This is the base on which the birthday attack operates. In informational security purposes, cryptographic hash functions exist where the data is taken and converted into a fixed-size bit string. To be ideal cryptographic hash functions, it must be easy to compute a hash value for a message, infeasible to create a message with a given hash, infeasible to modify a message without changing the hash, and infeasible to find different messages with the same hash. With this function, any tiny changes in the input will lead to a drastically different output. The goal for the attack is that for two different inputs x_1, x_2 , $f(x_1) = f(x_2)$. Now, the variable H is the number of values, and in the

birthday problem it was represented by 365. We can then use the formula from before to find the expected number of values before finding a collision.

$$\begin{aligned} p(n; H) &\approx 1 - e^{[-n(n-1)/(2H)]}. \\ &\approx 1 - e^{-n^2/(2H)}. \end{aligned}$$

Let $n(p; H)$ be the least number of values to choose, such that the probability for finding a collision is at least p .

$$\begin{aligned} n(p; H) &\approx \sqrt{2H \ln \frac{1}{1-p}}. \\ n(0.5; H) &\approx 1.1774\sqrt{H}. \end{aligned}$$

So the expected number of values that have to be chosen before finding the first collision is greater than $1.1774\sqrt{H}$.

In terms of actual use of this birthday attack, suppose Alice wants to trick Bob into signing a fraudulent contract. So, she creates both a fraudulent and fair contract. Then, on the fraudulent contract, she makes tiny edits, changing the contract by adding or deleting commas, having synonyms of the words, etc. Eventually, in theory, with enough fraudulent contracts, there will exist one which has the same hash value as the fair one. She then gets Bob's signature on the fair one and attaches it to the fake one, which both have the same hash value. However, this birthday attack can be made unfeasible by increasing the hash value output size until it is unfeasible to find a collision.

V. Conclusion

The birthday problem is an interesting model which shows the probability of collisions of 365 days in a year. While 365 different days seems like a large amount, only 23 people are required so that the probability is over 50%. There can be many different variations on the original problem, and many other problems can be solved using the same method as the birthday one. The birthday attack demonstrated this, as it showed the expected number of attempted values before finding a collision. In all, the birthday problem was an interesting problem to work on and makes me rethink the probability of all events.

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